

Overview of How DNS Works

Introduction

The Domain Name System (DNS) is a fundamental component of the Internet that provides the critical service of translating human-readable hostnames (e.g., `www.google.com`) into IP addresses (e.g., `142.250.190.14`). This translation is essential because while humans use hostnames, the underlying network protocols require IP addresses.

DNS as an Application-Layer Protocol

DNS is categorized as an application-layer protocol because:

1. It runs between communicating end systems using the client-server paradigm.
2. It relies on an underlying transport protocol (UDP, port 53) to exchange DNS messages.

Unlike other application-layer protocols (HTTP, FTP, SMTP), DNS is not directly used by end-users. Instead, it serves as a **core Internet function**, enabling applications to locate resources by resolving hostnames.

Why Not a Centralized DNS?

A centralized DNS design would be impractical due to:

- **Single Point of Failure:** Failure of the central server would disrupt the entire Internet.
- **Traffic Overload:** One server would be overwhelmed by billions of queries.
- **Geographical Distance:** Queries would experience long delays if the central server is far from the client.
- **Maintenance Issues:** Constant updates would be required for an extremely large database.

Conclusion: A centralized database does not scale. DNS adopts a **distributed, hierarchical design**.

Distributed, Hierarchical DNS Architecture

DNS uses a distributed architecture with multiple classes of servers:

Root DNS Servers

- Over 1000 instances worldwide, copies of 13 root servers.
- Managed by 12 organizations, coordinated by IANA.
- Provide IP addresses of Top-Level Domain (TLD) servers.

Top-Level Domain (TLD) Servers

- Handle domains such as .com, .org, .edu, and country codes (.uk, .jp).
- Maintained by organizations (e.g., Verisign for .com, Educause for .edu).
- Provide IP addresses of **authoritative DNS servers**.

Authoritative DNS Servers

- Store actual hostname-to-IP mappings for an organization.
- Organizations may run their own servers or outsource.
- Provide final answer in DNS query resolution.

Local DNS Servers

- Operated by ISPs or institutions.
- Act as intermediaries (proxies) between hosts and the DNS hierarchy.
- Often located near the client (on the same LAN or a few routers away).

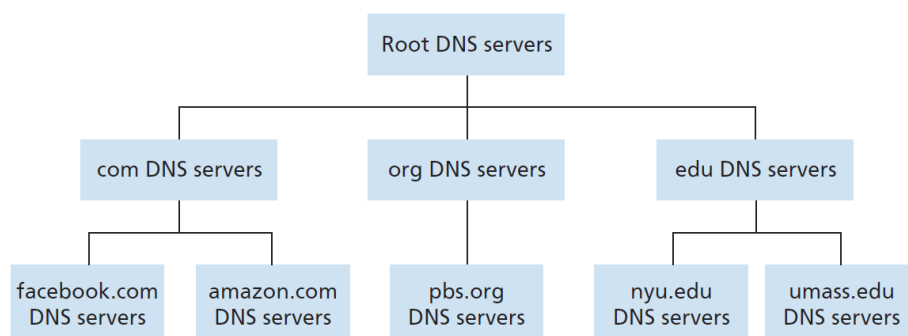


Figure 1: DNS Hierarchical Structure

DNS Query Resolution

Step-by-Step Resolution of `www.amazon.com`

The DNS query resolution process follows a hierarchy of servers:

1. The client contacts a **Root DNS Server**.
2. The Root server returns IP addresses of the **TLD servers** for `.com`.
3. The client queries a **TLD server**, which provides the IP address of the **Authoritative DNS server** for `amazon.com`.
4. The Authoritative server finally returns the IP address of `www.amazon.com`.

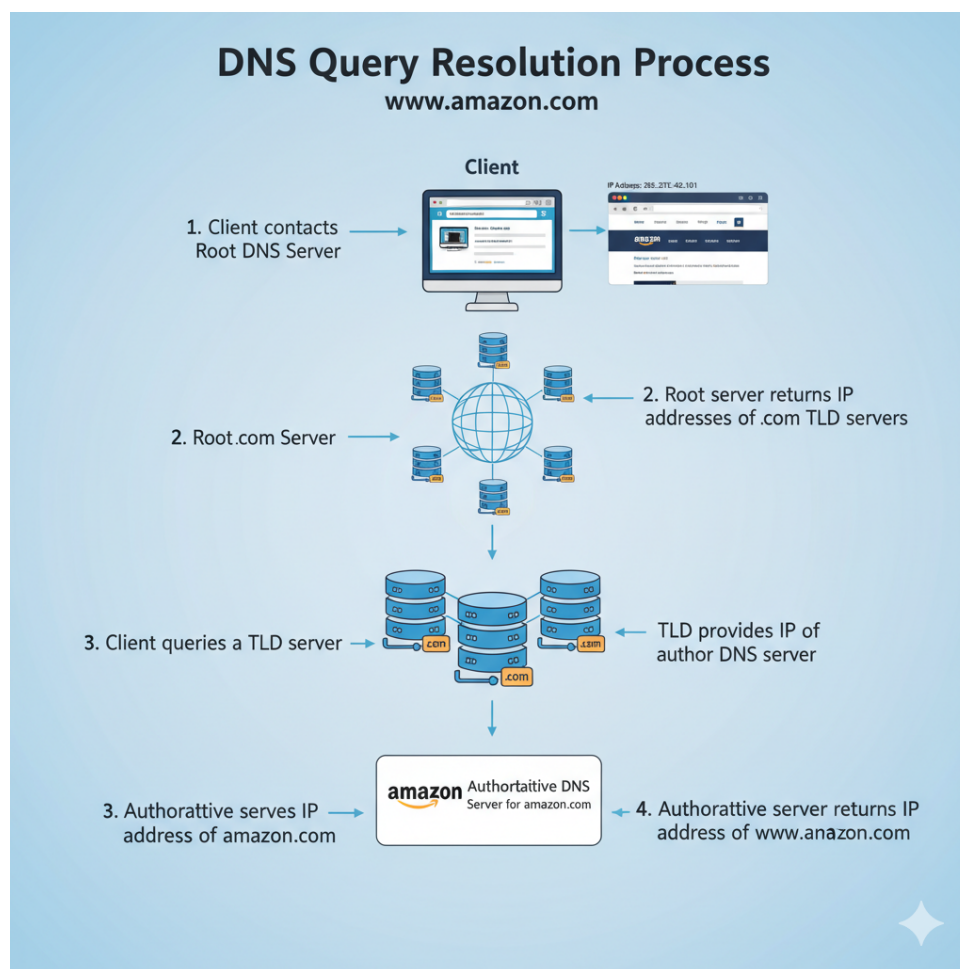


Figure 2: DNS Query Resolution Example for `www.amazon.com`

Role of the Local DNS Server

The Root, TLD, and Authoritative DNS servers belong to the **global DNS hierarchy** (Figure 1). However, there exists another key component: the **Local DNS Server**.

- Not strictly part of the hierarchy, but **central to DNS architecture**.

- Operated by ISPs (residential or institutional).
- Provided to hosts automatically (e.g., via DHCP).
- Acts as a **proxy** between the host and the DNS hierarchy.

Proximity:

- Institutional ISP: Local DNS often resides on the same LAN.
- Residential ISP: Typically separated by only a few routers.

Function: When a host issues a DNS query, the request is sent to the Local DNS Server, which then forwards the query into the DNS hierarchy and relays the response back to the host.

Detailed Example with Local DNS

Suppose the host `cse.nyu.edu` requests the IP address of `gaia.cs.umass.edu`.

- NYU's Local DNS server is `dns.nyu.edu`.
- An Authoritative DNS server for `gaia.cs.umass.edu` is `dns.umass.edu`.

Comparison of DNS Server Types

Server Type	Scope	Role in Resolution
Root Server	Global (13 logical servers, replicated worldwide)	Directs queries to the correct TLD servers.
TLD Server	Domain extensions (e.g., <code>.com</code> , <code>.org</code> , <code>.edu</code>)	Provides addresses of Authoritative DNS servers.
Authoritative Server	Organizational level (e.g., <code>amazon.com</code> , <code>umass.edu</code>)	Stores and returns the final hostname-to-IP mapping.
Local DNS Server	ISP-level (institutional or residential)	Acts as a proxy, forwarding queries into the hierarchy and caching results for efficiency.

Table 1: Comparison of Different DNS Server Types

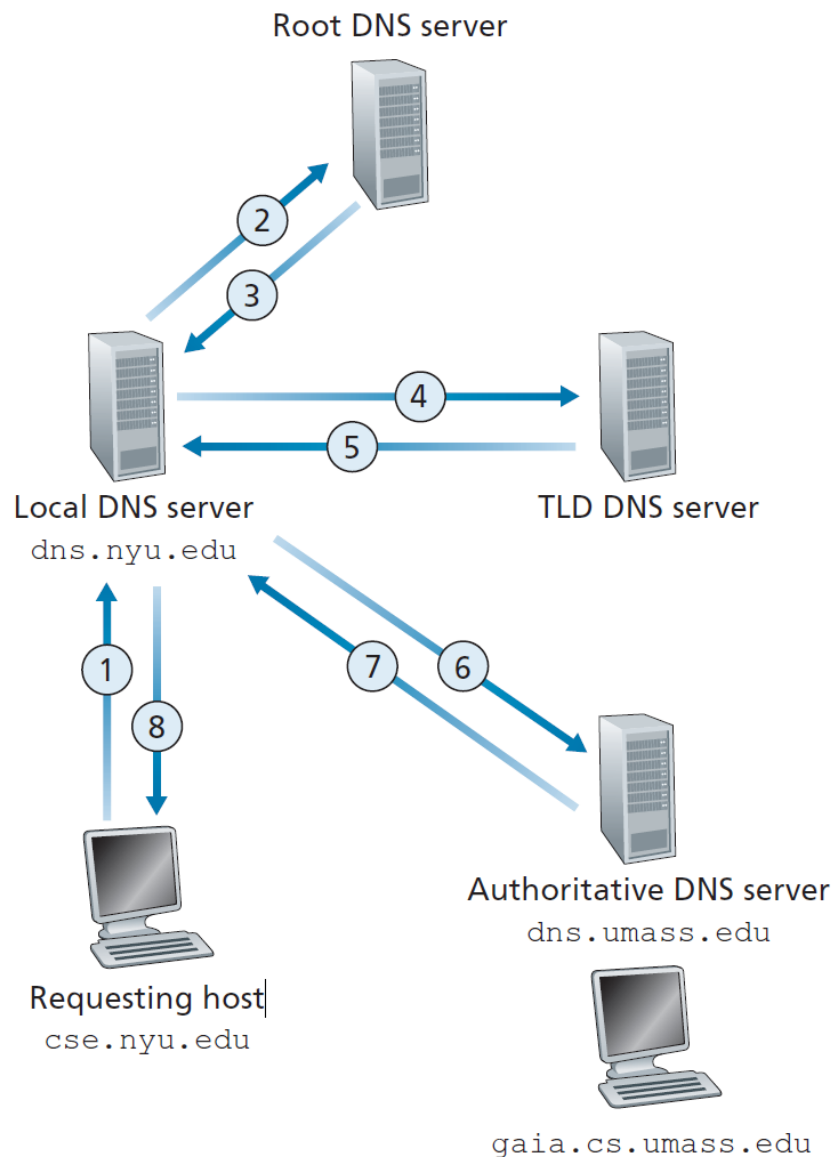


Figure 3: Interaction of Root, Local, TLD, and Authoritative DNS Servers for resolving `gaia.cs.umass.edu`

Recursive vs. Iterative Queries

- **Recursive query:** A server takes full responsibility to resolve the query on behalf of the client.
- **Iterative query:** A server returns the best answer it has (possibly another server's address), and the client continues the process.

Typically:

- Client → Local DNS server = Recursive.
- Local DNS server → Root/TLD/Authoritative = Iterative.

DNS Caching

- Improves efficiency and reduces query traffic.
- Servers cache responses (hostname-IP mappings, TLD addresses).
- Cached records have a **Time-To-Live (TTL)**, after which they expire.

Example: If `dns.nyu.edu` caches the IP of `cnn.com`, subsequent queries from NYU hosts (e.g., `kiwi.nyu.edu`) are answered instantly without contacting root/TLD servers.

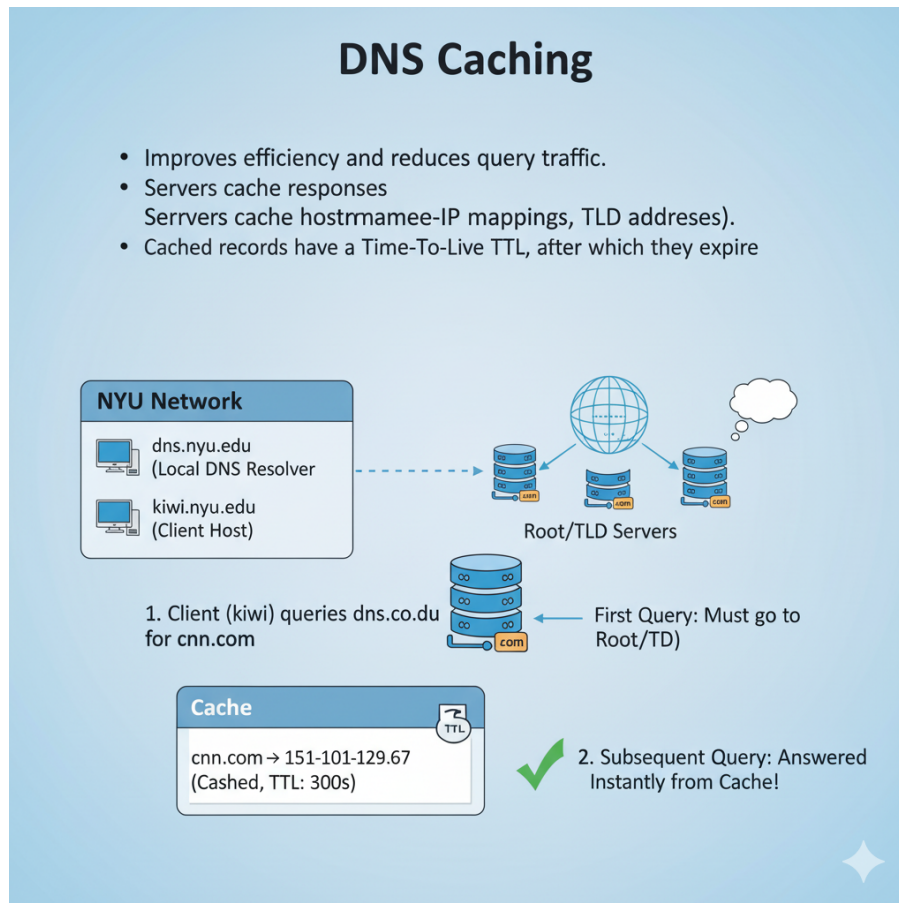


Figure 4: DNS Caching Reduces Load and Improves Efficiency

Key Takeaways

- DNS is a **distributed, hierarchical system** for hostname-to-IP translation.
- Root, TLD, and authoritative servers work together to resolve queries.
- Local DNS servers improve efficiency by caching results.
- Recursive and iterative queries combine to balance efficiency and scalability.
- Caching reduces load on root and TLD servers, making DNS scalable.